

■ AI Data Analyst - Executive Analysis Report

■ Executive Data Analysis Report

1. Executive Summary

- The dataset contains **136 students** across five (including two typo■variants) courses.
- **Data Science** dominates enrollment (**99 students**, ~73 % of the cohort).
- Gender distribution is skewed toward males (**99 M**, 72.8 % vs **37 F**, 27.2 %).
- Average student age is **22.2 years** (median = 22).
- Total payments amount to **\$150,860** with an average of **\$1,109.26** per student.
- **Data quality concerns:** 69 records missing *Enrollment_Date*, 21 duplicated *Student_IDs*, and numerous payment outliers identified via the IQR method.
- Recommendations focus on cleaning duplicate IDs, resolving missing enrollment dates, and investigating payment outliers before any predictive modeling or longitudinal analysis.

2. Dataset Overview & Data Quality Profile

| Aspect | Detail |

|-----|-----|

| **Raw shape** | 136 rows × 8 columns |

| **Columns** | *Student_ID*, *First_Name*, *Last_Name*, *Age*, *Gender*, *Course*, *Enrollment_Date*, *Total_Payments* |

| **Duplicates** | 0 rows (after deduplication) |

| **Missing values** | 69 missing *Enrollment_Date*; all other columns complete |

| **Data■type issues** | *Student_ID* and *Enrollment_Date* originally stored as strings; corrected during cleaning |

| **Quality flags** | - Duplicate *Student_IDs* (see anomalies)
 - Payment outliers (see anomalies) |

3. Data Cleaning & Validation Summary

| Step | Action | Outcome |

|-----|-----|-----|

| **Column split** | Parsed concatenated **Student_ID** field into 8 separate columns | All fields correctly separated |

| **Whitespace trim** | Stripped leading/trailing spaces from all object columns | Clean string values |

| **Type conversion** | **Age** → numeric, **Total_Payments** → float (removed \$/commas), **Enrollment_Date** → datetime | Proper dtypes for analysis |

| **Missing value imputation** | Numeric columns (**Age**, **Total_Payments**) → median; Categorical columns → mode | No NaNs in numeric/categorical fields |

| **Duplicate removal** | Dropped exact duplicate rows | 136 unique rows retained |

| **Validation** | Post-cleaning checks confirm 136 rows, 0 duplicates | Clean dataset ready for analysis |

4. User Query & Analytical Approach

User question:

> “Analyze key dataset metrics, profile anomalies, and generate summary insights.”

Analytical methodology (implemented in Pandas):

1. **Metric computation** – totals, counts, means, medians for demographics, age, payments, and course enrollment.

2. **Anomaly detection** –

- Missing value summary (already captured).
- Duplicate **Student_IDs** (identified via **duplicated**).
- Payment outliers using the IQR method.

3. **Insight generation** – textual bullet points derived from the computed metrics and anomalies.

All steps were executed in a single script (see Section 5) and stored in a dictionary **result** containing **metrics**, **anomalies**, and **insights**.

5. Python Analysis Code & Executed Data Output

Code

```
# Core summary metrics metrics = { "total_students": len(df), "gender_counts":
df["Gender"].value_counts().to_dict(), "average_age": df["Age"].mean(), "median_age":
df["Age"].median(), "course_counts": df["Course"].value_counts().to_dict(), "total_payments":
df["Total_Payments"].sum(), "average_payments": df["Total_Payments"].mean(),
"payments_by_course": ( df.groupby("Course")["Total_Payments"] .agg(["sum", "mean", "count"])
.reset_index() .to_dict(orient="records") ), "missing_values": df.isnull().sum().to_dict(), } #
Anomaly detection duplicate_mask = df["Student_ID"].duplicated(keep=False) duplicate_ids =
df.loc[duplicate_mask, "Student_ID"].unique().tolist() q1 = df["Total_Payments"].quantile(0.25)
```

```

q3 = df["Total_Payments"].quantile(0.75) iqr = q3 - q1 lower_bound = q1 - 1.5 * iqr upper_bound
= q3 + 1.5 * iqr outlier_mask = (df["Total_Payments"] < lower_bound) | (df["Total_Payments"] >
upper_bound) outlier_students = df.loc[outlier_mask, "Student_ID"].tolist() anomalies = {
"missing_values": metrics["missing_values"], "duplicate_student_ids": duplicate_ids,
"payment_outliers": outlier_students, } # Generate textual insights insights = [] if
metrics["course_counts"]: most_common_course = max(metrics["course_counts"],
key=metrics["course_counts"].get) insights.append( f"The most enrolled course is
**{most_common_course}** with " f"{metrics['course_counts'][most_common_course]} students." )
total = metrics["total_students"] male = metrics["gender_counts"].get("M", 0) female =
metrics["gender_counts"].get("F", 0) insights.append( f"Gender split: {male} male
({male/total:.1%}) and {female} female ({female/total:.1%})." ) insights.append( f"Overall
payments total ${metrics['total_payments']:.2f} with an average of "
f"${metrics['average_payments']:.2f} per student." ) enroll_missing =
metrics["missing_values"].get("Enrollment_Date", 0) if enroll_missing: insights.append(f"There
are {enroll_missing} records missing an enrollment date.") else: insights.append("All records
have an enrollment date.") if outlier_students: insights.append( f"Payment outliers detected
for Student_ID(s): {'', '.join(map(str, outlier_students))}." ) else: insights.append("No
payment outliers detected based on the IQR method.") # Compile final result result = {
"metrics": metrics, "anomalies": anomalies, "insights": insights, }

```

Executed Output (summarized)

Metrics

| Metric | Value |

|-----|-----|

| **Total students** | 136 |

| **Gender counts** | M: 99, F: 37 |

| **Average age** | 22.18 years |

| **Median age** | 22 years |

| **Course counts** | Data Science 99, Machine Learning 20, Web Development 13, Machine Learnin 3, Web Developmen 1 |

| **Total payments** | \$150,860 |

| **Average payment** | \$1,109.26 |

Payments by Course

| Course | Total (\$) | Avg per student (\$) | Count |

|-----|-----|-----|-----|

| Data Science | 106,710 | 1,077.88 | 99 |

| Machine Learning | 26,310 | 1,315.50 | 20 |

| Machine Learnin | 3,900 | 1,300.00 | 3 |

| Web Development | 12,990 | 999.23 | 13 |

| Web Developmen | 950 | 950.00 | 1 |

Anomalies

- **Missing values** – 69 missing **Enrollment_Date**.
- **Duplicate Student_IDs** – ['101', '102', '103', '104', '105', '106', '107', '108', '109', '110', '111', '112', '113', '114', '115', '', '500'] (21 unique IDs).
- **Payment outliers** – 71 IDs flagged (e.g., 101, 102, 103, 105, 106, 107, 108, 109, 111, 112, 113, 115, 500, 144, 157, 159, 217, 235, 237, plus several empty strings).

Insights (auto-generated)

1. The most enrolled course is **Data Science** with 99 students.
2. Gender split: 99 male (72.8 %) and 37 female (27.2 %).
3. Overall payments total **\$150,860.00** with an average of **\$1,109.26** per student.
4. There are **69 records missing an enrollment date**.
5. Payment outliers detected for many **Student_IDs** (see anomaly list).

6. Key Insights & Strategic Recommendations

■ Core Findings

- **Program focus:** The curriculum is heavily weighted toward *Data Science*. Consider expanding capacity in *Machine Learning* and *Web Development* to diversify offerings.
- **Gender imbalance:** Male students outnumber females by more than 2:1. Targeted outreach or scholarship programs could improve female participation.
- **Data completeness:** Over half of the records lack enrollment dates, limiting any cohort-based or temporal analysis (e.g., retention, seasonality).
- **Data integrity:** Duplicate IDs and a large set of payment outliers indicate upstream data-entry issues.

■ Recommendations

| Area | Action |

|-----|-----|

| **Data Quality** | • Deduplicate **Student_IDs** (investigate why IDs repeat).
• Resolve missing **Enrollment_Dates** – request source data or infer from related fields. |

| **Outlier Management** | • Review flagged payment outliers manually; verify if they are genuine high-value enrollments or entry errors.
• Apply robust statistical methods (e.g., Winsorization) if outliers are retained for modeling. |

| **Course Portfolio** | • Conduct a demand survey to gauge interest in expanding *Machine Learning* and *Web Development* tracks.
• Clean typo-variant course names (**Machine Learnin**, **Web Developmen**) to consolidate reporting. |

| **Gender Diversity** | • Launch mentorship or scholarship initiatives aimed at female applicants.
• Track gender trends over future enrollment cycles to assess impact. |

| **Future Analytics** | • Once enrollment dates are complete, perform cohort analysis (e.g., retention, payment timelines).
• Build predictive models for payment propensity after data■quality improvements. |

> **Bottom line:** The dataset provides a clear picture of current enrollment and payment patterns but is hampered by missing dates, duplicate identifiers, and numerous payment outliers. Addressing these data■quality gaps will unlock richer temporal insights and support more reliable strategic decision■making.